ASR-Style Transcript Normalization Removes Useful Pragmatic Signal for Dialogue Act Classification

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***Abstract*—Automatic speech recognition (ASR) pipelines often normalize transcripts by removing filled pauses and punctuation and expanding colloquial forms, potentially discarding cues useful for dialogue act classification. We evaluate nine transcript conditions across six classifiers (TF-IDF logistic regression, RoBERTa, BERT, ALBERT, GPT-2, and BiLSTM) and three conversational corpora (Switchboard, ICSI MRDA, and SLUE-HVB), supplemented by multi-seed experiments, normalization controls, and real-ASR evaluations. Across five random seeds, heavy normalization reduces Macro-F1 in all 15 neural model–corpus combinations under the original preprocessing protocols. Removal with the original filler/backchannel lexicon consistently degrades performance on MRDA and SLUE-HVB but produces mixed effects on Switchboard. Punctuation-aware TF-IDF experiments show that punctuation removal reduces Macro-F1 by 0.036 on Switchboard and 0.078 on MRDA, revealing that the original linear baseline's insensitivity was a tokenization artifact. Restricting filler removal to *um*, *uh*, and *erm* preserves degradation on MRDA but largely eliminates the SLUE-HVB effect, where the original lexicon removes class-defining backchannel expressions. These findings demonstrate that normalization effects depend on preprocessing definitions, feature extraction, and conversational context, supporting task-dependent rather than indiscriminate transcript normalization.**

***Index Terms*—automatic speech recognition, transcript normalization, dialogue act classification, disfluency, filled pauses**

I. Introduction

Automatic speech recognition (ASR) transcripts are often normalized by removing filled pauses, stripping punctuation, and standardizing colloquial expressions. Although these operations improve readability, they may discard cues relevant to turn management and discourse structure [1]–[3]. While previous studies have examined disfluencies and ASR errors, how normalization affects dialogue act classification across models and conversational settings remains insufficiently characterized.

We investigate nine transcript conditions across six classifiers (TF-IDF logistic regression, RoBERTa, BERT, ALBERT, GPT-2, and BiLSTM) and three corpora (Switchboard, ICSI MRDA, and SLUE-HVB), yielding 18 model–corpus combinations. The main experiments transform reference transcripts; separate analyses examine recognition errors in real ASR output. We also assess tokenization, lexicon sensitivity, and random-seed variability.

Our main contributions are:

1) A multi-model, cross-corpus evaluation showing consistent Macro-F1 degradation under heavy normalization across five random seeds, with more variable effects from filler removal.
2) A punctuation-aware TF-IDF analysis demonstrating that the apparent insensitivity of linear classifiers to punctuation removal results from tokenization rather than model architecture alone.
3) A restricted filler-lexicon analysis showing that degradation persists on MRDA but largely disappears on SLUE-HVB when class-defining backchannel tokens are preserved.
4) Cluster-level statistical analyses, real-ASR evaluations, and selective normalization experiments assessing the robustness and practical implications of transcript preprocessing.

Our findings show that normalization effects depend on the linguistic cues removed, feature extraction, and conversational setting, supporting task-dependent rather than indiscriminate transcript preprocessing.

II. Related Work

Disfluencies, filled pauses, and backchannels convey information about hesitation, turn management, and discourse structure [1]–[5]. Removing these expressions during transcript normalization may therefore eliminate cues useful for downstream pragmatic inference.

Prior studies have investigated ASR recognition errors and configurable transcription-noise evaluation [6]–[11]. However, recognition errors differ from deliberate transcript normalization, which can alter linguistic cues even when transcription is accurate.

Dialogue act recognition research has examined lexical and conversational cues [12]–[14]. Zelasko et al. [15] investigated the role of punctuation in transformer-based dialogue act recognition and segmentation.

TABLE I
DATASET STATISTICS AFTER RARE-LABEL FILTERING (≥ 50 OCCURRENCES).

	Switchboard	MRDA	HVB
Register	Telephone	Meetings	Call-center
Train/test	34.6k/9.1k	75.1k/16.7k	11.3k/6.1k
Labels	27	12	16
Canonical filler rate[a]	40.5%	15.9%	5.7%

[a] Utterances containing *um*, *uh*, or *erm*; the Switchboard rate is 17,703/43,763 utterances.

Disfluency studies establish the communicative role of hesitations and backchannels, whereas ASR-robustness work examines recognition errors and Zelasko et al. focus on punctuation in dialogue-act recognition. These studies do not jointly isolate deliberate filler/backchannel removal, punctuation stripping, and colloquial expansion for dialogue-act classification across model families and conversational registers. Our controlled text ablations address this gap with nine conditions, six classifiers, and three corpora, supplemented by tokenizer and lexicon controls, multi-seed evaluation, and real-ASR experiments.

III. EXPERIMENTAL SETUP

Datasets. We evaluate three conversational corpora: Switchboard [16], ICSI MRDA [17], [18], and SLUE-HVB [19], representing telephone conversations, multi-party meetings, and scripted call-center dialogues, respectively. SLUE-HVB is multi-label; approximately 40% of utterances have multiple tags, and we use the release's first tag as the primary class. Table I summarizes the datasets after removing labels with fewer than 50 instances in combined train–test counts. Switchboard uses a conversation-level 80/20 split (seed 42), while MRDA and HVB use their release-defined splits. The MRDA release describes its canonical partition as meeting-disjoint, although the reformatted records do not expose meeting identifiers.

Models. We evaluate TF-IDF with logistic regression (TF-IDF+LR), RoBERTa-base [20], BERT-base-cased [21], ALBERT-base-v2 [22], GPT-2 [23], and BiLSTM [24]. TF-IDF+LR uses unigram/bigram features ($\min_df = 3$, $\max_df = 0.95$) and class-balanced logistic regression. Transformers are fine-tuned for three epochs (learning rate 2×10^{-5}, batch size 16, maximum length 96). BiLSTM uses 200-dimensional embeddings, 128 hidden units, and 15 training epochs. Neural experiments use five seeds (1, 2, 3, 4, and 42); TF-IDF+LR is deterministic. We additionally evaluate a punctuation-aware, case-preserving TF-IDF tokenizer to examine punctuation discarded by the default tokenizer.

Normalization Conditions. We evaluate nine conditions: `gold_text` (original transcripts), `norm_colloq_only` (colloquial expansion), `rm_punct_only` (punctuation removal), `rm_discourse_only` (removal of *like*/*well*), `gold_alpha_text` (corpus-specific alphanumeric form), `rm_fillers_only` (filler/backchannel lexicon removal), `clean_light` (lowercasing and removal of non-alphanumeric characters except apostrophes), `clean_heavy` (light cleaning, filler removal, and colloquial expansion), and `asr_noise` (synthetic recognition errors).

Synthetic ASR noise applies token deletion (8%), substitution (8%), and insertion (4%). Each main-grid condition is applied to both training and test transcripts.

The original filler lexicon contains *um*, *uh*, *erm*, *mm*, *hmm*, *mhm*, *uhhuh*, *uh-huh*, and *huh*. The restricted lexicon retains only *um*, *uh*, and *erm*.

An additional `pragmatic_norm` control applies lowercasing and colloquial expansion while preserving fillers, backchannels, and punctuation. This transformation leaves MRDA unchanged; its reported neural comparisons are single-seed supplementary controls.

Normalization implementations have corpus-specific differences. Switchboard's original `clean_heavy` additionally removes *like* and *youknow*, while its `gold_alpha_text` uses ConvoKit's provided field, which also removes clitics. MRDA and HVB instead use regex-based alphanumeric filtering. Colloquial normalization changes no Switchboard or MRDA utterances, while punctuation removal and light cleaning leave HVB unchanged.

As a separate single-seed sensitivity test, we evaluate a harmonized strict heavy-normalization condition applying lowercasing, punctuation removal, colloquial expansion, and deletion of only *um*, *uh*, and *erm* across all three corpora.

Real-ASR Evaluation. We evaluate Whisper-base.en [25] under two protocols. On SLUE-HVB, models are trained and tested on ASR transcripts (17.7%/18.3% train/test WER) and compared with gold-trained/gold-tested models. On MRDA, gold-trained models are compared on gold and ASR test transcripts from a separately reconstructed 60/15 meeting-disjoint split (41.0% WER), combining recognition errors with train–test transcription mismatch. Switchboard uses synthetic ASR noise because its audio requires an LDC license.

Evaluation. Macro-F1 is the primary evaluation metric, assigning equal weight to each dialogue-act class. Primary neural results are summarized using mean and standard deviation across five seeds. At the primary seed (42), paired Macro-F1 differences are assessed using 2,000 cluster-bootstrap resamples, with conversations as clusters for Switchboard and calls for HVB. McNemar's test assesses paired utterance-level correctness rather than Macro-F1, with Bonferroni correction across 36 comparisons; 21 remain significant at $\alpha = 0.05/36$. Because utterances within conversations are dependent, McNemar results are supplementary to cluster-level uncertainty estimates. Individual p-values and complete condition-level outputs are included in the public repository.

Implementation scripts and supplementary materials are provided in a public project repository.[1]

IV. RESULTS

A. Multi-Seed Normalization Effects

Table II reports selected primary conditions across five seeds for the neural models and a deterministic TF-IDF+LR

[1] https://github.com/luckymaduabuchi/asr-normalization-dialogue-acts

TABLE II

MACRO-F1 UNDER ORIGINAL, FILLER-REMOVED, AND HEAVILY NORMALIZED TRANSCRIPTS. NEURAL RESULTS ARE MEAN ± STANDARD DEVIATION OVER FIVE SEEDS. TF-IDF+LR IS DETERMINISTIC. ORIGINAL CORPUS-SPECIFIC NORMALIZATION PROTOCOLS ARE USED.

Corpus	Model	Gold F1	Filler F1	Heavy F1
Switchboard	TF-IDF+LR	0.311	0.300	0.293
	RoBERTa	0.449 ± 0.006	0.447 ± 0.005	0.393 ± 0.012
	BERT	0.440 ± 0.005	0.449 ± 0.008	0.387 ± 0.004
	ALBERT	0.420 ± 0.008	0.409 ± 0.026	0.355 ± 0.046
	GPT-2	0.388 ± 0.012	0.383 ± 0.009	0.339 ± 0.008
	BiLSTM	0.370 ± 0.007	0.374 ± 0.012	0.336 ± 0.010
MRDA	TF-IDF+LR	0.409	0.364	0.361
	RoBERTa	0.593 ± 0.006	0.550 ± 0.003	0.485 ± 0.003
	BERT	0.591 ± 0.006	0.556 ± 0.002	0.481 ± 0.007
	ALBERT	0.580 ± 0.007	0.530 ± 0.016	0.469 ± 0.013
	GPT-2	0.565 ± 0.001	0.522 ± 0.008	0.444 ± 0.003
	BiLSTM	0.496 ± 0.005	0.438 ± 0.006	0.401 ± 0.011
SLUE-HVB	TF-IDF+LR	0.713	0.660	0.661
	RoBERTa	0.745 ± 0.008	0.697 ± 0.004	0.698 ± 0.004
	BERT	0.742 ± 0.004	0.698 ± 0.004	0.697 ± 0.003
	ALBERT	0.742 ± 0.004	0.697 ± 0.003	0.698 ± 0.003
	GPT-2	0.707 ± 0.026	0.671 ± 0.004	0.673 ± 0.007
	BiLSTM	0.732 ± 0.005	0.684 ± 0.008	0.683 ± 0.011

baseline. Under the original preprocessing protocols, heavy normalization reduces Macro-F1 in all 75 paired seed–model–corpus comparisons. Filler removal also reduces Macro-F1 in all 50 neural comparisons on MRDA and SLUE-HVB but produces mixed effects on Switchboard. For example, MRDA RoBERTa decreases from 0.593 to 0.550 after filler removal and to 0.485 under heavy normalization. In contrast, Switchboard BERT improves from 0.440 to 0.449 after filler removal but falls to 0.387 under heavy normalization. Thus, heavy normalization produces more consistent degradation than filler removal alone.

The complete primary-seed nine-condition grid is available in the repository. Across its 18 model–corpus combinations, synthetic ASR noise degrades all 18 ($\Delta = 0.033$–0.225). Punctuation removal and light cleaning each degrade 11 combinations and leave seven unchanged ($\Delta = 0$–0.084 for each operation). Colloquial expansion and discourse-marker removal have mixed, smaller effects ($|\Delta| \leq 0.011$); alphanumeric filtering is model-sensitive ($\Delta = -0.005$–0.209). The punctuation-aware control below explains the default TF-IDF result.

B. Punctuation and Filler-Lexicon Controls

The original TF-IDF+LR baseline appeared insensitive to punctuation removal because its default tokenizer discarded punctuation before feature extraction. Using a punctuation-aware, case-preserving tokenizer, punctuation removal reduces Macro-F1 by 0.036 on Switchboard and 0.078 on MRDA (Table III). Thus, the original near-neutral effect was a tokenization artifact rather than an intrinsic property of linear classifiers. SLUE-HVB remains unaffected because its transcripts lack punctuation.

We also evaluate a restricted filler lexicon containing only *um*, *uh*, and *erm*. On MRDA, filler-removal degradation remains positive across all six models ($\Delta = 0.015$–0.039), although the proportion of emptied backchannel utterances falls from 43.1% to 0.5%. Across the filtered SLUE-HVB data, the original lexicon empties 38 of 50 backchannel utterances, whereas the restricted lexicon empties none; only 16 backchannels occur in the held-out test set. Accordingly, filler-removal degradation approaches zero for the four evaluated neural models with non-anomalous primary-seed baselines ($\Delta = -0.006$ to 0.001).

TABLE III

PUNCTUATION-REMOVAL EFFECTS FOR TF-IDF+LR. Δ_{punct} DENOTES THE MACRO-F1 DECREASE RELATIVE TO THE CORRESPONDING GOLD CONDITION.

Corpus	Default Δ	Aware Gold	Aware Δ
Switchboard	0.0002	0.360	0.036
MRDA	0.0000	0.494	0.078
SLUE-HVB	0.0000	0.717	0.000

TABLE IV

PRIMARY-SEED MACRO-F1 DEGRADATION UNDER HARMONIZED STRICT HEAVY NORMALIZATION. $\Delta = \mathrm{gold} - \mathrm{strict}$; POSITIVE VALUES INDICATE DEGRADATION.

Model	SWB	MRDA	HVB
TF-IDF+LR	0.007	0.039	0.015
RoBERTa	0.044	0.080	−0.004
BERT	0.041	0.076	−0.001
ALBERT	0.032	0.077	0.003
GPT-2	0.048	0.110	−0.049
BiLSTM	0.022	0.056	0.000

Table IV reports results under an identical strict heavy-normalization function. All six models exhibit Macro-F1 degradation on Switchboard and MRDA, whereas five of six SLUE-HVB models change by at most 0.015. The exception is GPT-2, which improves by 0.049 from an unusually low gold baseline. All strict-control results use seed 42.

Separately, omitting Switchboard's extra *like*/*youknow* removal while retaining the nine-word lexicon changes the

TABLE V
SELECTED CLASS-LEVEL EFFECTS OF LEXICON-BASED REMOVAL FOR DETERMINISTIC TF-IDF+LR ON HELD-OUT TEST SPLITS. P, R, AND F1 DENOTE PRECISION, RECALL, AND F1-SCORE. BC DENOTES SLUE-HVB BACKCHANNELS.

Corpus	Class	Metric	Gold	No Fill.
Switchboard	b	P	0.558	0.436
	b	F1	0.433	0.365
MRDA	fh	R	0.495	0.330
	h	R	0.313	0.104
SLUE-HVB	BC	P	0.353	0.092
	BC	R	0.750	0.750
	BC	F1	0.480	0.163

normalized text for 17.7% of utterances. TF-IDF+LR Macro-F1 changes from 0.2933 to 0.2954; corresponding neural reruns are unavailable.

At seed 42, `pragmatic_norm` Macro-F1 remains within ± 0.013 of gold across 12 Switchboard and SLUE-HVB model combinations. It outperforms heavy normalization in 11 of 12 comparisons, suggesting that mild standardization may preserve performance when discourse cues remain.

C. Statistical and Class-Level Analysis

Five-seed standard deviations summarize training variability. Paired cluster-bootstrap confidence intervals quantify test-cluster sampling uncertainty at seed 42, not uncertainty in five-seed means. Throughout, positive Δ denotes the Macro-F1 decrease relative to gold.

At the primary seed, cluster-bootstrap 95% confidence intervals for heavy-normalization Macro-F1 differences exclude zero for all five neural models on Switchboard and four of five on SLUE-HVB. Filler-removal intervals exclude zero for only ALBERT on Switchboard and four neural models on SLUE-HVB. For TF-IDF+LR on Switchboard, filler-removal and heavy-normalization differences are 0.011 $[0.003, 0.020]$ and 0.017 $[0.009, 0.027]$, respectively. These are nominal 95% intervals without multiplicity adjustment; an F1 decrease alone does not establish statistical significance. The reformatted primary MRDA records lack meeting identifiers. On a separate exploratory 60/15 meeting-disjoint reconstruction, heavy-normalization intervals exclude zero for four of five neural models, while filler-only effects are split-sensitive.

Utterance-level McNemar tests provide an additional paired-correctness comparison but are not tests of Macro-F1 and do not adjust for conversation clustering; cluster-bootstrap intervals therefore remain the primary uncertainty evidence.

Table V shows different class-level effects across corpora. On Switchboard, backchannel precision decreases from 0.558 to 0.436, with an F1 reduction of 0.068. On MRDA, recall decreases for floor holders (`fh`) and holds before answers or agreements (`h`), indicating difficulty recognizing turn-management dialogue acts. In contrast, SLUE-HVB backchannel recall remains at 0.750, while precision decreases from 0.353 to 0.092, reducing F1 from 0.480 to 0.163. Because Macro-F1 weights this low-support class equally with the other 15 classes, its large F1 change warrants cautious interpretation. The restricted-lexicon analysis indicates that much of this degradation results from deleting class-defining backchannel expressions.

D. Real-ASR Evaluation

Whisper-base.en is evaluated under two protocols. On SLUE-HVB, models are trained and tested on Whisper transcripts (17.7%/18.3% train/test WER). Five models exhibit lower Macro-F1, whereas GPT-2 improves from 0.6674 to 0.6765 at seed 42. This isolated result does not establish an ASR benefit, especially given its unusually low baseline.

On MRDA, models trained on gold transcripts are evaluated on Whisper transcripts from 15 held-out meetings (WER: 41.0%). All six models degrade, with Macro-F1 decreases ranging from 0.1366 to 0.2822. Unlike SLUE-HVB, this experiment combines recognition errors with train–test transcription mismatch; the effect sizes are therefore not directly comparable.

V. DISCUSSION AND CONCLUSION

Our findings show that transcript normalization affects dialogue act classification differently across models, corpora, and preprocessing operations. Under the original protocols, heavy normalization reduces Macro-F1 across all five seeds for every neural model–corpus combination, whereas filler removal produces mixed effects on Switchboard but consistent degradation on MRDA and SLUE-HVB.

Interpretation of normalization effects. The punctuation-aware TF-IDF control shows that punctuation is predictive when the feature extractor preserves it. The default TF-IDF baseline appeared insensitive because its tokenizer discarded punctuation. Filler-lexicon controls reveal a different mechanism: on MRDA, degradation persists when removal is restricted to *um*, *uh*, and *erm*, suggesting that these expressions provide useful classification cues. On SLUE-HVB, most degradation disappears under the restricted lexicon because the original inventory removes class-defining backchannel expressions. Thus, not all performance losses reflect the same type of information removal.

Robustness and practical implications. The five-seed experiments show that heavy-normalization degradation is more consistent than filler-removal effects. Under harmonized strict normalization, degradation persists across all six classifiers on Switchboard and MRDA but is largely absent on SLUE-HVB, demonstrating sensitivity to corpus characteristics and lexicon definitions. Single-seed pragmatic-normalization controls further suggest that modest text standardization can retain dialogue-act performance when backchannels and punctuation are preserved.

Limitations. The study is limited to three English-language corpora. SLUE-HVB findings concern a first-tag, single-label reduction and do not evaluate its full multi-label dialogue-act task. The original Switchboard heavy-normalization implementation additionally removes *like* and *youknow*, and harmonized lexicon experiments use a single seed. On Switchboard,

spaced backchannel expressions such as *Uh - huh* are partially affected by the strict lexicon, which removes *uh* but retains *huh*. Thus, identical normalization rules do not necessarily remove equivalent linguistic cues across corpora. According to the release-defined canonical split, the primary MRDA partition is meeting-disjoint but not speaker-disjoint; because meeting identifiers are absent from the reformatted records, meeting-level confidence intervals are unavailable for that primary split. The alternative MRDA neural comparisons use different training scripts, hyperparameters, and text formatting, preventing controlled attribution of their differences to normalization. Real-ASR evaluations use different protocols: matched ASR training/testing on SLUE-HVB and gold-trained/ASR-tested models on MRDA. The pragmatics-preserving control uses reference transcripts and does not recover cues already missing from real ASR output. All reported BiLSTM results use the corrected empty-input pooling implementation.

Overall, aggressive transcript normalization can degrade dialogue act classification, but the magnitude and mechanism depend on feature extraction, lexicon design, and conversational setting. These findings support selective, task-dependent normalization rather than indiscriminate transcript cleaning. Future work will examine multi-seed harmonized normalization, matched ASR training on MRDA, and additional conversational domains.

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